

EE 446 Lab 5

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GitHub Link: <https://github.com/zelamirs-hub/EE-446-TinyML/tree/Lab-5>

1. Converting the audio file to a spectrogram/MFCC creates a 2D image of the word's frequency features. Then 2D convolution can be used to extract features to be fed into the neural network.
2. The representative dataset is used to determine the range of integers mapped from the float32 values. This helps the model recognize typical inputs which increases accuracy.
3. `audio_processor.get_data()` obtains and processes audio signals from datasets. The parameters are presented below.

```
audio_processor.get_data(HOW_MANY, ----- # of items to fetch
                        OFFSET, ----- index of item
                        MODEL_SETTINGS, ----- dictionary of configurations
                        BACKGROUND_FREQUENCY, ----- frequency added of noise
                        BACKGROUND_VOLUME_RANGE, ----- volume of noise
                        TIME_SHIFT_MS, ----- max time shift in ms
                        MODE, ----- dataset to pull: train/test
                        SESS) -----Active TensorFlow Session
```

4. The `g_model` is the array of raw hexadecimal bytes that make up the model. The `g_model_len` is the total amount of bytes the model contains, since each element is an `int8`, each element is one byte. Therefore the `g_model_len` is also equal to the number of elements in the array.

micro_speech_lab_5 | Arduino IDE 2.3.10

File Edit Sketch Tools Help

Arduino Nano 33 BLE

micro_speech_lab_5.ino arduino_audio_provider.cpp arduino_command_responder.cpp arduino_main.cpp audio_provider.h command_responder.h ...

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Output Serial Monitor X

Message (Enter to send message to 'Arduino Nano 33 BLE' on 'COM20') Both NL & CR 115200 baud

Heard unknown (203) @23440ms
Heard yes (212) @25552ms
Heard yes (208) @27120ms
Heard no (205) @32464ms
Heard no (202) @34112ms
Heard unknown (202) @39552ms
Heard unknown (219) @46544ms
Heard no (210) @48944ms
Heard unknown (206) @52176ms
Heard no (204) @54016ms
Heard unknown (201) @57968ms
Heard unknown (209) @60928ms
Heard yes (212) @62848ms
Heard yes (207) @69392ms
Heard unknown (204) @70784ms
Heard unknown (209) @72336ms
Heard unknown (203) @74464ms
Heard unknown (203) @76016ms

Ln 1, Col 1 Arduino Nano 33 BLE on COM20 2